

Virtual Landscapes Geological Mapping

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This activity was selected for the On the Cutting Edge Reviewed Teaching Collection

This activity has received positive reviews in a peer review process involving five review categories. The five categories included in the process are

- Scientific Accuracy
- Alignment of Learning Goals, Activities, and Assessments
- Pedagogic Effectiveness
- Robustness (usability and dependability of all components)
- Completeness of the ActivitySheet web page

For more information about the peer review process itself, please see https://serc.carleton.edu/teachearth/activity_review.html.

- First Publication: January 22, 2018
- Reviewed: December 10, 2020 -- Reviewed by the On the Cutting Edge Activity Review Process

Summary

A virtual landscape is a virtual reality environment designed to develop geologic mapping and field skills. They include natural landscapes where the geology can be mapped, and interactive 3D block models of topographic and geological maps. The worlds can be found at www.sec.leeds.ac.uk/virtual-landscapes/.

Topics

[Geology](#)

Grade Level

College Upper (15-16), Graduate/Professional

Readiness for Online Use

[Online Ready](#)



Follow the link above to find [activities](#) from Teach the Earth on a specific topic.

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Context

Audience

Virtual landscapes are designed to be used in class/lab activities, but can also be used independently online. The simpler landscapes are designed for upper college, the more complex are undergraduate level.

Skills and concepts that students must have mastered

The concept that the type of rocks seen across a landscape can vary in a systematic way that can be recorded to produce a map. Guidance will be needed in the different mapping techniques, what symbols to use etc as these are not directly covered on the site. All readings are given in strike/dip/dip direction.

How the activity is situated in the course

There are multiple activities that can be centred round the virtual landscapes. The 3D topographic map can be used to introduce students to the concept of reading the landscape from a map. The 3D geological maps allow students to observe how changes in the strike or dip of a rock change the outcrop pattern and can be used to encourage students to develop their own rules for how patterns change. The mapping landscapes can be used before going on the first geological mapping field trip and as part of the geological maps course.

For other Teach the Earth pages using Virtual Landscapes see References and Resources below.

Goals

Content/concepts goals for this activity

Skills developed include how to plot field data, understand the different methods of field mapping, how to create a geological map. They develop 3D visualisation skills and provide accessible alternatives to fieldwork for those unable to go in the field.

Higher order thinking skills goals for this activity

Understanding the 3D nature of geological units and how these are shown on 2D geologic maps. How to extrapolate isolated and spatially distributed data to create a map.

Skills goals for this activity

Other skills include producing a geological map and associated cross sections and stratigraphic columns. The data can also be written up in the form of a field report.

Description and Teaching Materials

The activity is online at www.see.leeds.ac.uk/virtual-landscapes/ The landscapes can be played directly online or downloaded for offline use. We recommend downloading the landscapes as this gets round slow internet connections etc. The accompanying field slips are available for download. Ideas for exercises and activities for the different landscapes are in the following document together with some background information:

- [Virtual Landscapes Notes.docx](#) (Microsoft Word 2007 (.docx) 27kB Apr29 20)

Solutions to the exercises and field slips with outcrop localities marked on them are available via the TeacherStash.

- Download all files as a zipped file: [revirtualworldsinformationonsercwebsite.zip](#) (a 12.4MB Zip Archive file) educator-only file, or individually:
 - [01_Topo map detailed.pdf](#) (a 659kB Acrobat (PDF) file) educator-only file
 - [02_Strike-maps.pdf](#) (a 1.4MB Acrobat (PDF) file) educator-only file
 - [02_Dip-maps.pdf](#) (a 1MB Acrobat (PDF) file) educator-only file
 - [03_Lighthouse Bay answer map.pdf](#) (a 408kB Acrobat (PDF) file) educator-only file
 - [03_Lighthouse Bay field slip no outcrops.pdf](#) (a 373kB Acrobat (PDF) file) educator-only file
 - [03_Lighthouse Bay field slip with outcrops.pdf](#) (a 390kB Acrobat (PDF) file) educator-only file
 - [04_Rhoscolyn mapping slip no outcrops.pdf](#) (a 6.6MB Acrobat (PDF) file) educator-only file
 - [04_Rhoscolyn mapping slip with outcrops.pdf](#) (a 464kB Acrobat (PDF) file) educator-only file
 - [04_Rhoscolyn whole area no outcrops.pdf](#) (a 497kB Acrobat (PDF) file) educator-only file
 - [04_Rhoscolyn whole area with outcrops.pdf](#) (a 470kB Acrobat (PDF) file) educator-only file
 - [04_Rhoscolyn answer map.pdf](#) (a 7.9MB Acrobat (PDF) file) educator-only file
 - [05_Three_River_Hills_Fieldslip.pdf](#) (a 1.8MB Acrobat (PDF) file) educator-only file
 - [05_Three River Hills section answer.pdf](#) (a 164kB Acrobat (PDF) file) educator-only file
 - [05_Three River Hills map answer.pdf](#) (a 1.9MB Acrobat (PDF) file) educator-only file

Teaching Notes and Tips

Students tend to rush round the mapping landscapes collecting data as fast as possible without thinking about interpretation. We encourage students to slow down and plan their routes across the landscape in a time efficient manner. We strongly emphasise the need to understand the geology as they work and to draw boundaries as they go along and not treat it as a "join the dots" exercise. We suggest they work a transect first to get an overall idea of the geology and then map the boundaries.

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Assessment

The goal of the mapping landscapes is to construct a geological map and cross section. Completion of this and comparison with the answer allows students to see what they have done correctly and where they have gone wrong. This can be used as either formative or summative assessment.

References and Resources

Website with the landscapes, field slips and instructions is www.see.leeds.ac.uk/virtual-landscapes/

Other pages using the Virtual Landscapes: https://serc.carleton.edu/NAGTWorkshops/online_field/activities/236670.html and https://serc.carleton.edu/NAGTWorkshops/online_field/activities/237061.html

Houghton JJ; Lloyd GE; Robinson A; Gordon CE; Morgan DJ (2015) The Virtual Worlds Project: Geological mapping and field skills, *Geology Today*, 31, pp.227-231. doi: 10.1111/gto.12117